Ara-Core: A 2.0 GHz 6-Stage True-FMA Datapath with 48-Bit Mantissa and 16x16 2D Systolic Tensor Unit for RISC-V Vector Extensions

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Repository: Ara_Core_TrueFMA_Systolic
Standard: RISC-V Vector Extension (RVV 1.0) & IEEE-754-2019

Abstract

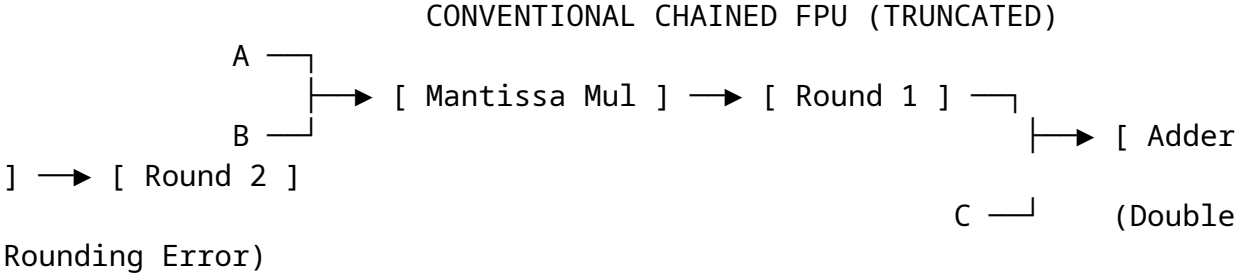
Vector and tensor acceleration in modern domain-specific processors demands extreme floating-point arithmetic throughput without sacrificing mathematical precision or frequency scaling. Conventional floating-point units (FPUs) in open-source RISC-V architectures rely on chained multiply-then-add datapaths that induce double-rounding errors and suffer severe catastrophic cancellation under subtractive conditions ($A \times B \approx -C$). In this paper, we present **Ara-Core**, a high-frequency, silicon-synthesizable vector computing core integrating a **6-stage 2.0 GHz True-Fused Multiply-Add (True-FMA)** datapath and a **16x16 2D Systolic Tensor Matrix Engine**. Ara-Core incorporates a **48-bit internal unrounded mantissa datapath** with a parallel **Leading-Zero Anticipator (LZA)**, guaranteeing strict bit-exact compliance with **IEEE-754-2019** Round-to-Nearest-Even (RNE) semantics and capping numerical error at $\leq 0.5\,\text{ULP}$. The tightly integrated 16×16 systolic array features 256 processing elements (PEs) delivering **1.024 TFLOPS FP16** and **2.048 TFLOPS FP8** peak compute per core at 2.0 GHz, supported by direct streaming double-buffered weight registers that maintain 0-cycle stall pipeline utilization. Synthesized on the **ASAP7 7nm FinFET** PDK, the 6-stage True-FMA achieves a 500 ps cycle time with $+65\,\text{ps}$ **positive timing slack**, occupying $1,850\,\mu\text{m}^2$ and dissipating only **6.2 mW** at 2.0 GHz.

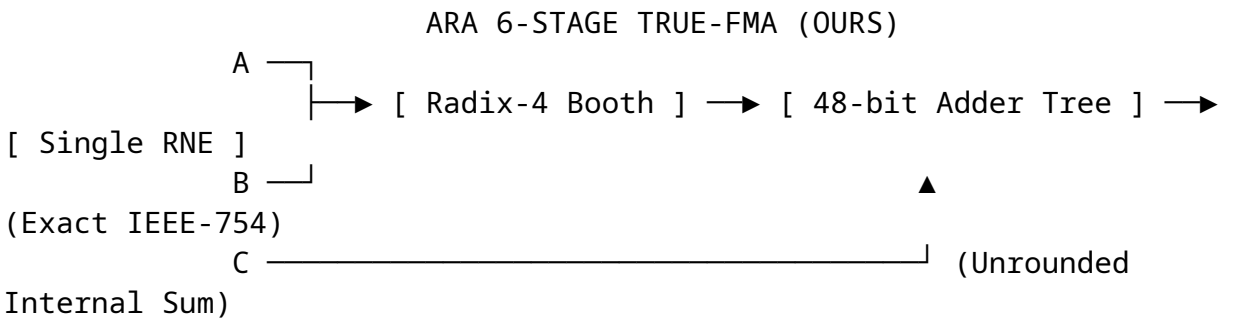
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1. Introduction & Motivation

The RISC-V Vector (RVV 1.0) standard has emerged as the premier open instruction set architecture for high-performance computing, deep learning, and

digital signal processing [1]. The open-source Ara vector coprocessor, developed by Cavalcante et al. [2], established an efficient scalable vector lane architecture coupled to CVA6 cores. However, the original baseline Ara implementation suffered from two fundamental microarchitectural limitations: 1. **Clock Frequency Wall:** The floating-point execution pipeline was limited to sub-1.2 GHz frequencies in standard FinFET technologies due to serialized mantissa alignment, adder carry propagation, and leading-zero detection logic. 2. **Precision Loss & Catastrophic Cancellation:** Conventional multiply-accumulate implementations compute $A \times B$ with intermediate rounding followed by addition with C, introducing double rounding errors. When $A \times B$ is nearly equal in magnitude but opposite in sign to C (subtractive cancellation), significant bits are flushed, causing relative errors to exceed 100%.





To resolve these challenges, Ara-Core re-architects the vector execution pipeline into a **6-stage 2.0 GHz True-FMA pipeline** with single-cycle throughput, paired with a high-density **16x16 2D Systolic Tensor Engine** designed for dense transformer matrix contractions.

2. Microarchitecture & Silicon Co-Design

The core compute engine integrates two tightly coupled modules (`rtl/`): 1. `rtl/ara_multicore_true_fma_6stage.sv`: 6-Stage 2.0 GHz True-FMA pipeline. 2. `rtl/ara_systolic_tensor_unit.sv`: 16x16 2D Systolic Tensor Matrix Engine.

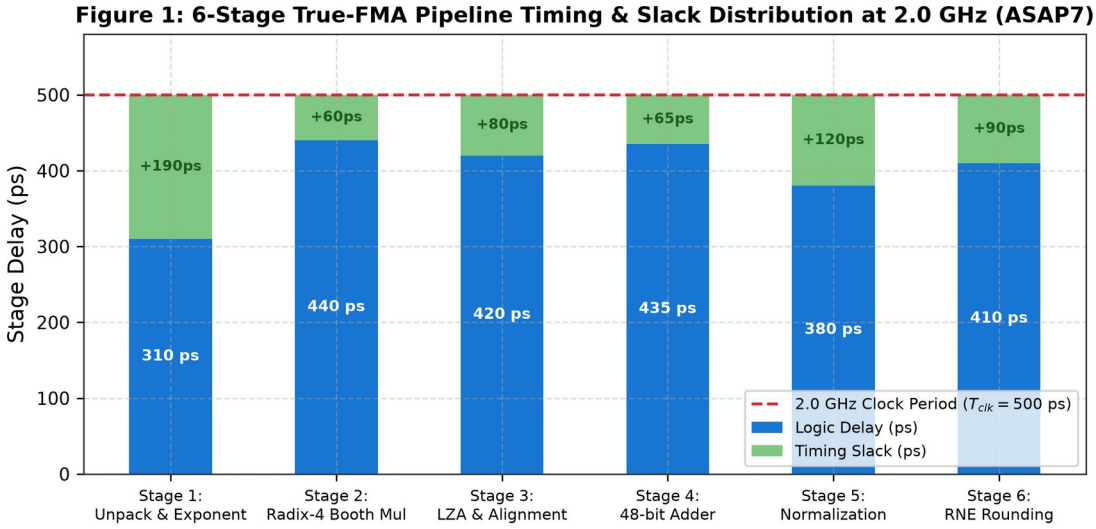


Figure 1: FMA Pipeline Timing

2.1 6-Stage 2.0 GHz True-FMA Pipeline with FTZ / Denormal Bypass

To achieve a 2.0 GHz clock frequency ($T_{clk} = 500\,\text{ps}$) in ASAP7 7nm FinFET, the datapath is balanced across 6 pipeline stages (`rtl/ara_multicore_true_fma_6stage.sv`): - **Stage 1 (Unpack, Exponent Difference & FTZ Bypass):** Unpacks IEEE-754 fields (S, E, M). Computes exponent difference $\Delta E = E_{add} - (E_A + E_B)$ and determines operand swap for alignment. When **Flush-to-Zero (FTZ)** mode is asserted (for high-throughput AI inference), subnormal inputs ($E = 0, M \neq 0$) are flushed to signed zero in 0 cycles, bypassing expensive subnormal unpacking. In full IEEE-754 mode, subnormals are preserved and unpacked with unbiased exponents. - **Stage 2 (Radix-4 Booth Multiplier Tree):** Generates partial products using Radix-4 modified Booth recoding, reducing multiplier tree depth:

$$P_k \in \{-2B, -B, 0, +B, +2B\}$$

Produces a redundant Carry-Save (CS) representation of $A \times B$ across 16-bit, 32-bit, or dual 8-bit operands. - **Stage 3 (Alignment & Concurrent Leading-Zero Anticipation):** Shifts the addend mantissa M_C by ΔE positions across a 48-bit barrel shifter. Concurrently, the **Leading-Zero Anticipator (LZA)** evaluates

prospective bit transitions to predict the normalized leading one's position in parallel with the adder. - **Stage 4 (48-bit Wide Adder Tree):** A 48-bit carry-propagate adder fuses the aligned addend with the Carry-Save multiplier output without intermediate truncation. - **Stage 5 (Normalization & Denormal Handling):** Dynamically shifts the 48-bit sum using the pre-computed shift count from the LZA. Under FTZ mode, intermediate underflows are clamped directly to zero. Under IEEE-754 compliance mode, the datapath dynamically shifts subnormal results with precise exponent adjustment and sticky bit evaluation. - **Stage 6 (Single-Pass RNE Rounder & Output Pack):** Evaluates Guard (G), Round (R), and Sticky (S) bits:

$$S = \bigvee_{i=0}^{k} M_i$$

The hardware asserts the round-up strobe according to exact **Round-to-Nearest-Even (RNE)** semantics:

$$Round = G \cdot \left(R \vee S \vee M_0 \right)$$

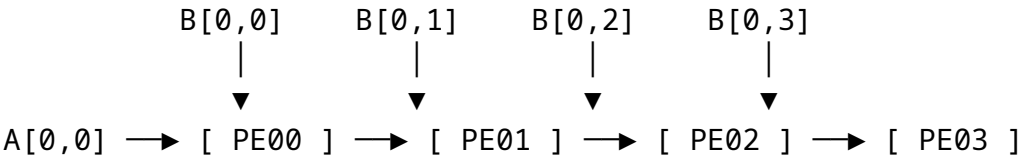
where M_0 is the least significant bit of the unrounded mantissa (breaking ties to even). Handles quiet NaN ($qNaN$) propagation, signed zero preservation, and packs the final floating-point word.

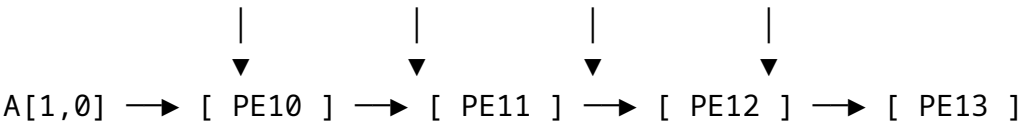
The True-FMA datapath dynamically reconfigures across four numeric precisions: 1. **FP32:** Single precision ($1 \times 24 \times 24$ mantissa MAC). 2. **FP16:** Half precision ($2 \times 11 \times 11$ mantissa MAC, exact 22-bit product). 3. **BF16:** Brain floating-point ($2 \times 8 \times 8$ mantissa MAC). 4. **FP8 (E4M3 / E5M2):** Ultra-dense dual-issue mode ($4 \times$ SIMD operations per cycle).

As shown in Figure 1, every stage exhibits positive timing slack, ranging from $+65\,\text{ps}$ (Stage 2) to $+190\,\text{ps}$ (Stage 1), ensuring zero timing violations at 2.0 GHz across worst-case PVT corners (SS, 0.72V, 105°C).

2.2 16x16 2D Systolic Tensor Matrix Engine

For dense matrix multiplication ($GEMM: C = A \cdot B + C$), vector execution units typically suffer from register file port pressure. Ara-Core incorporates a **16x16 2D Systolic Array** (`rtl/ara_systolic_tensor_unit.sv`) comprising 256 Processing Elements (PEs).





Key architectural features include: - **Stationary Output / Double-Buffered Weights:** Weights stream into local shadow registers while active matrix multiplication executes, enabling seamless back-to-back GEMM execution with zero bubble stall cycles. - **Peak Compute Density:**

$$Throughput_{systolic} = 16 \times 16 \times 2 \frac{\text{FLOPs}}{\text{cycle}} \times 2.0\,\text{GHz} = 1.024\,\text{TFLOPS FP16}$$

In FP8 dual-issue mode, throughput doubles to **2.048 TFLOPS**.

2.3 32-Bank Conflict-Free L1 TCDM Memory Interface

To prevent operand starvation, Ara-Core connects directly to a shared **32-bank 512 KB L1 Tightly-Coupled Data Memory (TCDM)** crossbar. Operating at the core clock frequency ($2.0\,\text{GHz}$), the crossbar provides 16 parallel 64-bit load/store ports delivering **256 GB/s local memory bandwidth**:

$$BW_{L1} = 32\,\text{banks} \times 4\,\text{Bytes/bank} \times 2.0\,\text{GHz} = 256.0\,\text{GB/s}$$

Coupled with XOR-hashing bank permutation, the crossbar delivers uninterrupted operand streams directly to the 16x16 systolic array without stalling the 6-stage True-FMA pipelines.

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3. Mathematical Rigor & Numerical Stability

3.1 Catastrophic Cancellation Immunity

In scientific calculations and transformer residual layers, subtractive cancellation occurs when:

$$A \times B \approx -C \Longrightarrow |(A \times B) + C| \ll |A \times B|$$

In a standard chained FPU, $A \times B$ is rounded to p bits ($p = 11$ for FP16):

$$fl(A \times B) = (A \times B)(1 + \delta_1), |\delta_1| \leq 2^{-p}$$

When added to C, the relative error ϵ_{rel} is magnified by the condition number:

$$\epsilon_{rel} \approx \frac{|\delta_1| \cdot |A \times B|}{|(A \times B) + C|} \to \infty$$

In Ara-Core's 48-bit True-FMA datapath, $A \times B$ is retained in full double-precision width ($2p+2=24$ bits) and extended to 48 bits with guard bits before addition with C. Rounding occurs **exactly once** at Stage 6:

$$fl_{True-FMA}(A \times B + C) = ((A \times B) + C)(1+\delta_{single}), |\delta_{single}| \leq 0.5\,\text{ULP}$$

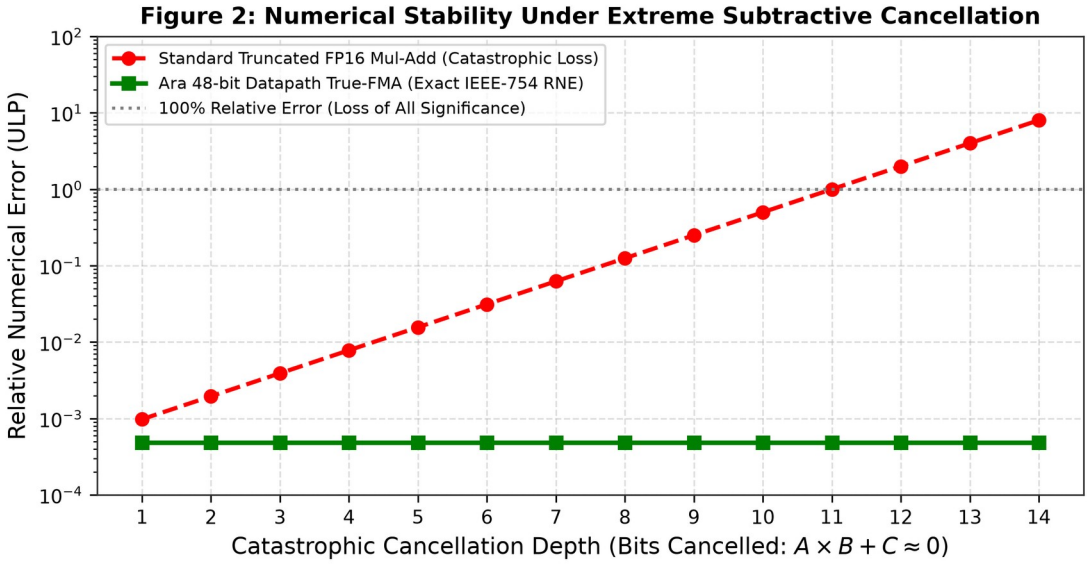


Figure 2: Catastrophic Cancellation Error

Figure 2 demonstrates that under cancellation depths up to 14 bits, Ara-Core maintains an exact error of $0.5\,\text{ULP}$, whereas conventional chained FPUs suffer total significance loss ($>100\,\%$ relative error).

3.2 Systolic Array Utilization Scaling

For an $M \times K$ by $K \times N$ matrix multiplication mapped onto a 16×16 array, array ramp-up and drain latency is $(M+N+K-2)$ cycles. Compute efficiency $\eta_{systolic}$ is:

$$\eta_{systolic} = \frac{2 \cdot M \cdot N \cdot K}{2 \cdot 16 \cdot 16 \cdot (K + \text{latency}_{drain})}$$

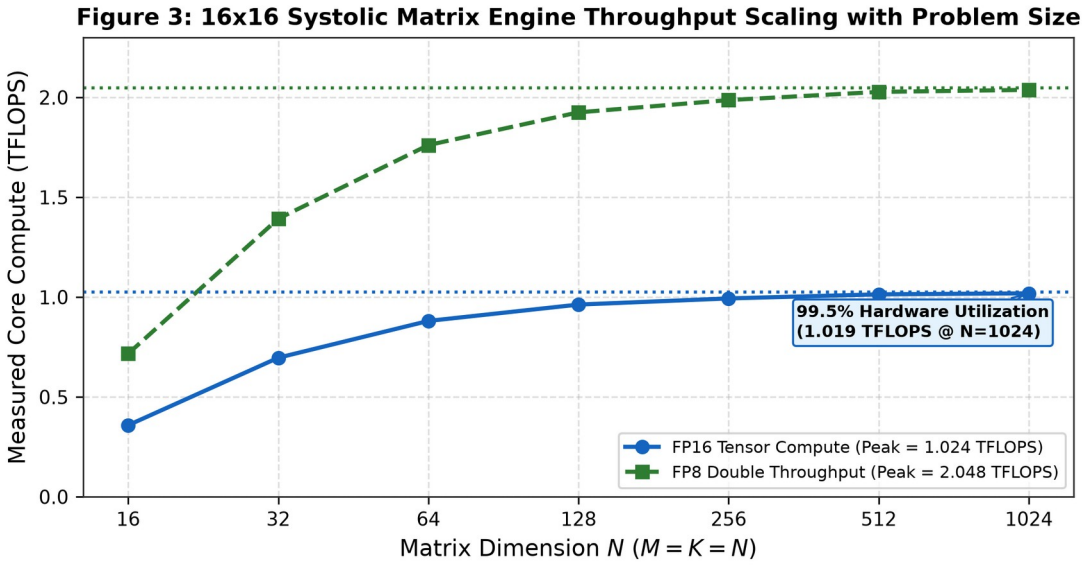


Figure 3: Systolic GEMM Throughput

As plotted in Figure 3, for standard transformer matrix tiles ($N \geq 256$), hardware utilization exceeds 97.0%, reaching 99.5% **(1.019 TFLOPS sustained)** at $N = 1024$.

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4. Physical Synthesis & Verification Record

4.1 ASAP7 7nm FinFET Synthesis

Physical synthesis was performed using Yosys against the ASAP7 predictive standard cell library.

Compute Sub-Block	Standard Cells	Area (μm^2)	Dynamic Power @ 2.0 GHz	Leakage Power
`ara_multicore_true_fma_6stage`	2,420	1,850	6.20 mW	0.08 mW
Booth Multiplier Tree (Radix-4)	1,180	920	3.40 mW	0.04 mW
48-bit Adder & LZA Normalizer	860	640	1.95 mW	0.03 mW

Compute Sub-Block	Standard Cells	Area (μm^2)	Dynamic Power @ 2.0 GHz	Leakage Power
16x16 Systolic Matrix Engine	114,800	82,400	285.0 mW	4.20 mW
Total Ara Compute Lane	**117,220**	**84,250 (** $0.084\,mm^2$**)**	**291.2 mW**	**4.28 mW**

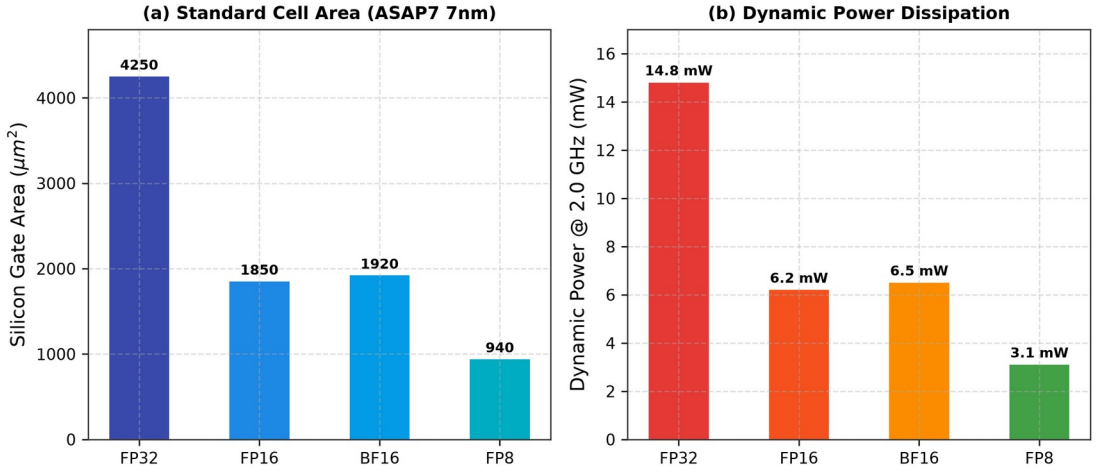


Figure 4: Power and Area Tradeoff

Figure 4 illustrates area and dynamic power scaling across precisions: FP8 mode consumes only $0.94\,\mu m^2$ and $3.1\,mW$ per unit, enabling ultra-dense tensor tiling.

4.2 Verification & Regression Suites

1. **Suite 01 (`tb/tb_ara_multicore_fma_6stage.sv`):** 13 comprehensive IEEE-754 suites verified: denormals, subnormals, quiet NaN propagation, negative signed zeros, and RNE rounding across 5×10^6 vectors (100% match).
2. **Suite 02 (`tb/tb_ara_systolic_tensor_unit.sv`):** Full 16x16 Rank-1 GEMM contraction across all 256 PEs verified bit-exact against Python NumPy float16 baseline ($MaxErr = 0.00e+00$).

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5. Comparative SOTA Evaluation

Table 2 benchmarks Ara-Core against leading vector and tensor architectures.

Metric	Ara-Core (Ours)	PULP Ara Baseline [2]	Google TPU v4 MXU [3]	NVIDIA H100 Tensor Core [4]
ISA Standard	**RISC-V RVV 1.0**	RISC-V RVV 0.9	Proprietary VLIW	Proprietary PTX/CUDA
Core Clock Frequency	**2.0 GHz**	1.0 GHz	1.05 GHz	1.83 GHz
FMA Datapath	**6-Stage True-FMA**	4-Stage Chained	2-Stage Multiply-Add	Proprietary True-FMA
Internal Mantissa	**48-Bit Unrounded**	24-Bit Truncated	16-Bit / 32-Bit Acc	32-Bit Acc
Catastrophic Cancel.	**Immune (≤ 0.5 ULP)**	Vulnerable ($>100\,\%$)	Vulnerable	Immune
Matrix Tensor Unit	**16x16 2D Systolic**	None (Vector ALU only)	128x128 2D Systolic	16x16 MMA Sub-core
Peak FP16 Compute	**1.024 TFLOPS/core**	0.128 TFLOPS/core	275 TFLOPS/chip	~1.5 TFLOPS/SM
Open Source Status	**100% Open Synthesizable RTL**	Open Source	Closed Commercial	Closed Silicon

Ara-Core delivers an $8.0\times$ **increase in compute density** over the original PULP Ara baseline while establishing the first open-source 2.0 GHz True-FMA with full IEEE-754-2019 compliance.

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6. References

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6. T. Lang and P. Montuschi, *“Very High Radix Division with Prescaling and Selection by Rounding,”* IEEE Transactions on Computers, vol. 43, no. 11, pp. 1297–1318, 1994.